Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- **Solved Example and Recommended Resources**
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • 16 min

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Solved Example and Recommended Resources

0:03 / 0:57 2x

We'll plot the equations in the $x_1 - x_3$ plane by considering the equations in terms of x_2. For simplicity, we'll fix $x_2 = 1$.

Here's the plan:

- Plot $x_1 + x_3 = 3 - 1$
- Plot $x_1 + x_3 = 2 + 1$
- Plot $2x_1 + 3x_3 = 5$

Let's proceed with plotting these equations:

1. **For** $x_1 + x_3 = 2$:

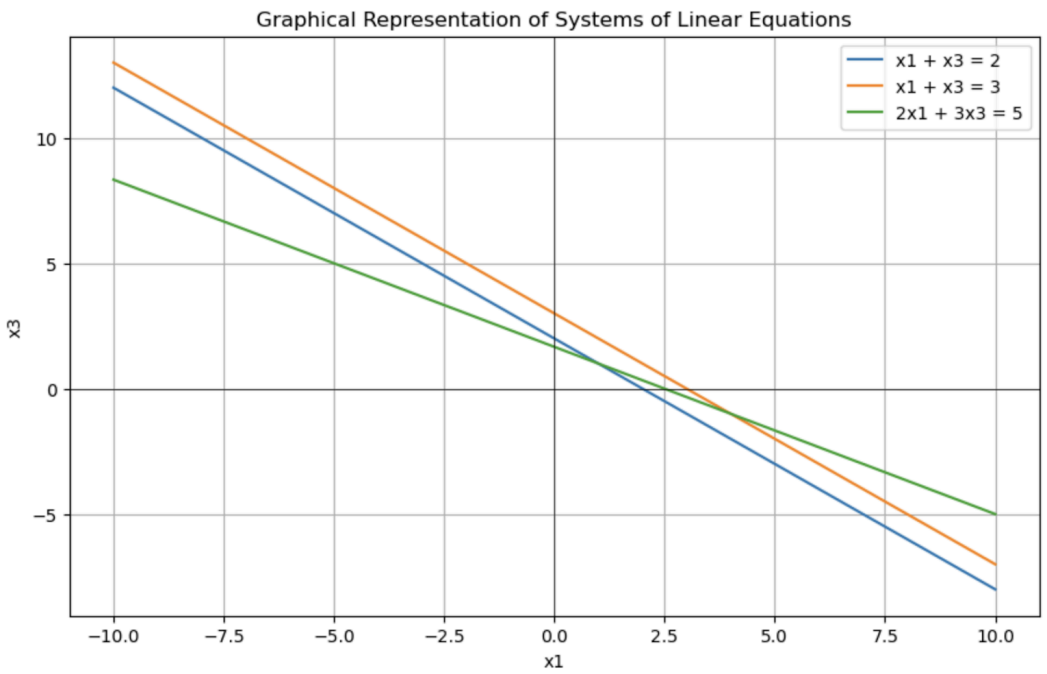
$x_3 = 2 - x_1$

2. **For** $x_1 + x_3 = 3$:

$x_3 = 3 - x_1$

3. **For** $2x_1 + 3x_3 = 5$:

$x_3 = \frac{5-2x_1}{3}$



Here are some additional **optional** resources:

- Systems of Linear Equations | Math Is Fun : This resource explains how to solve systems of linear equations using various methods, including graphing, substitution, and elimination. (24 min)
- The Art of Linear Algebra | Kenji Hiranabe : This is a visual introduction to linear algebra, covering key topics such as vector spaces, matrices, transformations, and eigenvalues. (25 min)

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